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United States Patent	12413815
Kind Code	B2
Date of Patent	September 09, 2025
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Information processing device and information processing method

Abstract

An information processing device (**100**) according to an aspect of the present disclosure includes a control unit (**160**) configured to perform a process of acquiring content selection information about a user; a process of determining a display content and content selection candidate information based on the content selection information; and a process of outputting the display content and a first display image corresponding to the selection candidate information of the content.

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Appl. No.:	18/041647
Filed (or PCT Filed):	August 18, 2021
PCT No.:	PCT/JP2021/030145
PCT Pub. No.:	WO2022/044904
PCT Pub. Date:	March 03, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20230328319 A1	Oct. 12, 2023

Foreign Application Priority Data

JP	2020-145069	Aug. 28, 2020
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Publication Classification

Int. Cl.: H04N21/45 (20110101); H04N21/431 (20110101); H04N21/482 (20110101)

U.S. Cl.:

CPC H04N21/4532 (20130101); H04N21/4312 (20130101); H04N21/482 (20130101);

Field of Classification Search

CPC: H04N (21/4512); H04N (21/4532); H04N (21/482)

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a U.S. National Phase of International Patent Application No. PCT/JP2021/030145 filed on Aug. 18, 2021, which claims priority benefit of Japanese Patent Application No. JP 2020-145069 filed in the Japan Patent Office on Aug. 28, 2020. Each of the above-referenced applications is hereby incorporated herein by reference in its entirety.

FIELD

(2) The present disclosure relates to an information processing device and an information

processing method.

BACKGROUND

(3) Currently, there are various types of broadcasting such as terrestrial broadcasting, satellite broadcasting, cable television broadcasting, and Internet broadcasting, and the number of channels is as large as several hundred channels. In an area where the large number of channels can be viewed, that is, a multi-channel area, in order to select a content that the user wants to view on a television or the like, for example, the user performs an operation of frequently switching channels, an operation of selecting a channel that the user wants to view from an electronic program guide (EPG), or the like.

(4) However, the operation as described above is likely to be a complicated operation, and it is difficult for the user to select a channel to be viewed in a multi-channel area. In order to facilitate this channel selection, a technique has been proposed in which, when a television is powered on, a channel having the highest viewing frequency is selected among channels selected in the past time zone corresponding to the time when the television is powered on (see, for example, Patent Literature 1).

CITATION LIST

Patent Literature

(5) Patent Literature 1: JP 2010-263510 A

SUMMARY

Technical Problem

(6) However, in the above-described conventional technology, a channel desired to be viewed by the user, that is, a content desired by the user is not necessarily displayed. In a case where the content that the user desires to view is not displayed, a complicated operation is required to select the content that the user desires to view, and thus, it is difficult to easily view the content that the user desires.

(7) Therefore, the present disclosure proposes an information processing device and an information processing method capable of easily viewing a content desired by a user.

Solution to Problem

(8) According to the present, an information processing device including a control unit configured to perform a process of acquiring content selection information about a user, a process of determining a display content and content selection candidate information based on the content selection information, and a process of outputting the display content and a first display image corresponding to the selection candidate information is provided.

(9) According to the present, an information processing method executed by a computer, the method including a process of acquiring content selection information about a user, a process of determining a display content and content selection candidate information based on the content selection information, and a process of outputting the display content and a first display image corresponding to the selection candidate information is provided.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram for describing an overview of an information processing device according to the first embodiment.

(2) FIG. 2 is a block diagram illustrating a hardware configuration of the information processing device according to the first embodiment.

(3) FIG. 3 is a block diagram illustrating a configuration of a logical function of the information processing device according to the first embodiment.

(4) FIG. 4 is a diagram for describing a first display screen at the time of selecting a usual filter

according to the first embodiment.

(5) FIG. 5 is a diagram for describing a second display screen (usual program guide) at the time of selecting a usual filter according to the first embodiment.

(6) FIG. 6 is a first diagram for describing priority at the time of selecting a usual filter according to the first embodiment.

(7) FIG. 7 is a second diagram for describing priority at the time of selecting a usual filter according to the first embodiment.

(8) FIG. 8 is a flowchart illustrating a flow of a first display process at the time of selecting a usual filter according to the first embodiment.

(9) FIG. 9 is a flowchart illustrating a flow of a second display process at the time of selecting a usual filter according to the first embodiment.

(10) FIG. 10 is a first diagram for describing a usual filter setting screen according to the first embodiment.

(11) FIG. 11 is a second diagram for describing a usual filter setting screen according to the first embodiment.

(12) FIG. 12 is a diagram for describing a first display screen at the time of selecting a genre filter according to a second embodiment.

(13) FIG. 13 is a first diagram for describing a second display screen (preference program guide) at the time of selecting a genre filter according to the second embodiment.

(14) FIGS. 14A, 14B, 14C, and 14D are first diagrams for describing priority at the time of selecting a genre filter according to the second embodiment.

(15) FIG. 15 is a second diagram for describing priority at the time of selecting a genre filter according to the second embodiment.

(16) FIG. 16 is a flowchart illustrating a flow of a first display process at the time of selecting a genre filter according to the second embodiment.

(17) FIG. 17 is a second diagram for describing a second display screen (preference program guide) at the time of selecting a genre filter according to the second embodiment.

(18) FIG. 18 is a diagram for describing a selectable display of a usual filter according to another embodiment.

(19) FIG. 19 is a first diagram for describing customization of a usual program guide according to another embodiment.

(20) FIG. 20 is a second diagram for describing customization of a usual program guide according to another embodiment.

(21) FIG. 21 is a third diagram for describing customization of a usual program guide according to another embodiment.

(22) FIG. 22 is a first diagram for describing a highlight display of a TV menu according to another embodiment.

(23) FIG. 23 is a second diagram for describing a highlight display of a TV menu according to another embodiment.

DESCRIPTION OF EMBODIMENTS

(24) Hereinafter, the embodiments of the present disclosure will be described in detail with reference to the drawings. The drawings illustrate only an example. In the following embodiments, the same parts are denoted by the same reference signs, and a duplicate description will be omitted.

(25) The description will be given in the following order. 1. First Embodiment 1-1. Basic configuration example 1-2. Functional configuration example 1-3. Display screen at the time of selecting usual filter (TV menu, usual program guide) 1-4. Priority at the time of selecting usual filter 1-5. First display process (TV menu) at the time of selecting usual filter 1-6. Second display process at the time of selecting usual filter (usual program guide) 1-7. Third display process (voice input operation) at the time of selecting usual filter 1-8. Usual filter setting screen 1-9. Summary of first embodiment 2. Second Embodiment 2-1. Display screen at the time of selecting genre filter

(TV menu and preference program guide) 2-2. Priority at the time of selecting genre filter 2-3. First display process (TV menu) at the time of selecting genre filter 2-4. Second display process (preference program guide) at the time of selecting genre filter 2-5. Recommendation display of preference program guide 2-6. Summary of second embodiment 3. Other embodiments 3-1. Various settings by control panel 3-2. Selectable display of usual filter 3-3. Customization of usual program guide 3-4. Highlight display of TV menu 4. supplementary note

1. First Embodiment

1-1. Basic Configuration Example

(26) A basic configuration of an information processing device according to the first embodiment will be described with reference to FIGS. 1 and 2. FIG. 1 is a diagram for describing an overview of the information processing device according to the first embodiment. FIG. 2 is a block diagram illustrating a hardware configuration of the information processing device according to the first embodiment.

(27) As illustrated in FIG. 1, an information processing device **100** functions as a receiving device that receives broadcast signals broadcast from a broadcasting tower **200** by respective broadcasting station and content data distributed from the Internet **220** (specifically, a content distribution server communicatively connected via the Internet). The information processing device **100** is a television device including a display screen (display). On the display screen, for example, as illustrated in FIG. 1, an image G1 of the display content is displayed on the entire screen (the entire region of the screen), and a TV menu (mini program guide) G2 which is a user interface image (UI image) is displayed in the lower region of the screen.

(28) Note that the information processing device **100** is not limited to a television device, but may be an external device (external equipment) such as a smartphone, a projector, a car navigation system, a computer, a set top box (STB), a recorder having a record and playback function, or a player having a playback function. For example, a set top box, a recorder, and a player are used by being connected to a television device.

(29) As illustrated in FIG. 2, the information processing device **100** includes a camera **101**, a microphone **102**, an input device **103**, a communication interface (I/F) **104**, a memory **105**, a tuner **106**, a decoder **107**, a display **108**, a speaker **109**, a remote control I/F **110**, a bus **111**, and a processor **112**.

(30) The camera **101** includes an imaging element such as a charge coupled device (CCD) or a complementary metal oxide semiconductor (CMOS), and captures an image. An image (for example, each frame constituting a moving image) captured by the camera **101** is treated as an input image for processing by the information processing device **100**.

(31) The microphone **102** collects a voice uttered by the user and generates a voice signal. The voice signal generated by the microphone **102** is treated as an input voice for voice recognition by the information processing device **100**.

(32) The input device **103** is a device used by the user to directly operate the information processing device **100**. For example, the input device **103** includes buttons, switches, dials, and the like disposed in a housing of the information processing device **100**. When detecting an input operation (user input) of the user, the input device **103** generates an input signal corresponding to the detected input operation by the user.

(33) The communication I/F **104** mediates communication between the information processing device **100** and another device (external device). For example, the communication I/F **104** supports any wireless communication protocol or wired communication protocol, and establishes a communication connection with another device. Furthermore, the communication I/F **104** receives a content signal from an external device (for example, an Internet distribution server), and extracts content data from the content signal.

(34) Note that the communication I/F **104** includes a communication device or the like for connecting it to a network. The network connected to the communication I/F **104** is a network

connected in a wired or wireless manner, and is, for example, the Internet, a home local area network (LAN), or the like.

(35) For example, the communication I/F **104** is a communication card or the like for wired or wireless LAN, Bluetooth (registered trademark), Wi-Fi (registered trademark), or wireless USB (WUSB). Furthermore, the communication I/F **104** may be a connection port such as a Universal Serial Bus (USB) terminal, an optical audio terminal, or a high-definition multimedia interface (HDMI) (registered trademark) terminal. Furthermore, the communication I/F **104** may be, for example, a router for optical communication, a router for asymmetric digital subscriber line (ADSL), a modem for various communications, or the like. The communication I/F **104** transmits and receives signals and the like to and from the Internet and other communication devices using a predetermined protocol such as TCP/IP.

(36) The memory **105** includes a storage medium such as a semiconductor memory or a hard disk, and stores a program and data for processing by the information processing device **100**, and content data. The data stored in the memory **105** includes, for example, data necessary for various processes described later. Note that some or all of the programs and data may be acquired from an external data source (for example, a data server, a network storage, an external memory, or the like) without being stored in the memory **105**.

(37) The tuner **106** extracts and demodulates a content signal of a desired channel from a broadcast signal received via an antenna (not illustrated). Then, the tuner **106** outputs the demodulated content signal to the decoder **107**.

(38) The decoder **107** decodes content data from a content signal input from the tuner **106**. Furthermore, the decoder **107** may decode content data from a content signal received via the communication I/F **104**. A content image is generated based on the content data decoded by the decoder **107**.

(39) The display **108** includes a screen (display screen) including a liquid crystal display (LCD), an organic light-emitting diode (OLED), a cathode ray tube (CRT), or the like. The display **108** displays an image generated by the information processing device **100** on a screen.

(40) The speaker **109** includes circuit elements such as a diaphragm and an amplifier to output a voice based on an output voice signal generated by the information processing device **100**. The volume of the speaker **109** can be changed according to an input operation by the user to the input device **103** or an input operation by the user to a local terminal such as a remote controller.

(41) The remote control I/F **110** receives a remote control signal (for example, infrared signals or other wireless signals) transmitted from a local terminal such as a remote controller used by a user. When detecting the remote control signal, the remote control I/F **110** generates an input signal corresponding to the detected remote control signal. In addition, examples of the local terminal may include a smartphone, a tablet terminal, or a personal computer (PC) in addition to the remote controller.

(42) The bus **111** mutually connects the camera **101**, the microphone **102**, the input device **103**, the communication I/F **104**, the memory **105**, the tuner **106**, the decoder **107**, the display **108**, the speaker **109**, the remote control I/F **110**, and the processor **112**.

(43) The processor **112** is, for example, a central processing unit (CPU), a digital signal processor (DSP), or the like. The processor **112** realizes various functions of the information processing device **100** to be described later by executing a program stored in the memory **105** or another storage medium.

(44) The basic configuration of the information processing device **100** according to the first embodiment is described above. However, the hardware configuration illustrated in FIG. 2 is an example, and the present invention is not limited to this. For example, the information processing device **100** may not include the camera **101**, and the microphone **102** may be provided in an external device (for example, a voice terminal such as a smart speaker or a mobile terminal such as a smartphone) connected to the information processing device **100** or an input terminal such as a

remote controller.

(45) Note that the information processing device **100** is a device capable of receiving video and voice data from an external device such as a recorder, a game machine, a PC, a digital camera, or a smartphone connected via the communication I/F **104**, and outputting and playing back the data from the display **108** or the speaker **109**. Furthermore, a plurality of external devices can be connected to the information processing device **100** (alternatively, a plurality of external devices can be connected to the information processing device **100** via a distributor). In a situation where one or more external devices are connected to the information processing device **100**, the user can select or switch the external device by the user's input operation.

1-2. Functional Configuration Example

(46) Next, a functional configuration of the information processing device according to the first embodiment will be described with reference to FIG. **3**. FIG. **3** is a block diagram illustrating a configuration of a logical function of the information processing device according to the first embodiment.

(47) As illustrated in FIG. **3**, the information processing device **100** includes a voice acquisition unit **120**, a feature database (DB) **140**, a control unit **160**, a history DB **172**, and a preference DB **174**. The control unit **160** includes a voice recognition unit **130**, an application unit **150**, a search control unit **162**, a selection control unit **164**, and a display control unit **166**.

(48) Note that some of the functional blocks illustrated in FIG. **3** may be realized in an external device (for example, a device in a cloud computing environment) of the information processing device **100**. Furthermore, for example, the control unit **160** may cause an external control function to execute the processing (control process) described below instead of executing the processing by itself.

(49) The voice acquisition unit **120** acquires a voice signal generated by the microphone **102** as an input voice. Then, the voice acquisition unit **120** outputs the acquired input voice to the voice recognition unit **130**.

(50) The voice recognition unit **130** recognizes the voice of the user based on the input voice input from the voice acquisition unit **120**. For example, the voice recognition unit **130** converts the input voice into a text indicating the content thereof to output the text to the control unit **160** or the application unit **150**. Note that the start process of voice reception and voice recognition may be automatically started, or may be started using a button or an icon disposed in a local terminal such as a remote controller.

(51) For example, the voice recognition unit **130** may identify the speaker (user) by using the voice feature data stored in the feature DB **140** in recognition of the input voice input from the voice acquisition unit **120**. The voice recognition unit **130** outputs the identification result of the speaker to the control unit **160** and the application unit **150**.

(52) Note that, in the first embodiment, the information processing device **100** performs a voice recognition process, but the present invention is not limited to this. For example, the voice recognition process may be performed by an external device connected to the information processing device **100**. Examples of the external device include a local terminal such as a smart speaker, a smartphone, a tablet terminal, or a PC, a server on the Internet, or the like.

(53) The feature DB **140** stores in advance voice feature data used for voice recognition by the voice recognition unit **130**. The voice feature data includes, for example, a voice feature amount indicating a feature of utterance for each user.

(54) Note that the feature DB **140** may store image feature data used for image recognition. The image feature data includes, for example, a known image feature amount of a predetermined portion such as a hand or a face of the user. The image feature data may include face image data for each user. Furthermore, the feature DB **140** may be constructed in the information processing device **100** or may be constructed on an external device such as an external server.

(55) The history DB **172** stores history information about a viewing history and a reservation

history of the contents (broadcast program, recorded program, Internet distributed program, application, and the like). The viewing history includes a viewing time and a viewing time duration of the content, a usage count and a usage time (application selection information) of the application, user information, a search history by the user, a selection history of the search result, and the like. The reservation history includes a content reservation time, a content reserved viewing time, and the like. The history information is generated for each user (for example, a father, a mother, a son, and the like), and is stored in the history DB **172** in association with each user. Such history information is included in content selection information (content selection record information) used for search for and determination of the content viewed by the user.

(56) Note that the history information is generated for each user, but the present invention is not limited to this, and the information may not be generated for each user. Furthermore, the history information is obtained by detecting the viewing history or the reservation history of the user by the information processing device **100**, but the present invention is not limited to this. For example, the history information may be acquired from various external devices and set in the history DB **172**. Furthermore, the history DB **172** may be constructed in the information processing device **100** or may be constructed on an external device such as an external server.

(57) The preference DB **174** stores preference information about the user. The preference information includes, for example, information about a sport, a sports team, an entertainer, a singer, a writer, an original author of a drama, a character, and the like that the user likes. As in the history information, the preference information is generated for each user (for example, a father, a mother, a son, and the like) and stored in the history DB **172** in association with each user. Such history information is included in content selection information used for search for and determination of content viewed by the user.

(58) Note that the preference information is generated for each user, but the present invention is not limited to this, and may not be generated for each user. Furthermore, the preference information is obtained by detecting the user's preference (for example, which content the user has frequently viewed, and the like) based on the history information or the like, but the present invention is not limited to this. For example, the preference information may be set in the preference DB **174** in advance by a user's input operation or the like. Furthermore, the preference DB **174** may be constructed in the information processing device **100** or may be constructed on an external device such as an external server.

(59) The application unit **150** executes various application functions of the information processing device **100**. For example, a program playback function, an electronic program guide display function, a recording setting function, an external input switching function, a photograph regeneration function, a moving image playback function, a music playback function and an Internet browsing function, an image recognition function, and the like are executed by the application unit **150**. The application unit **150** outputs an application image (including a content image) and a voice generated through the application function to the control unit **160**. Note that at least part of the application function cooperates with the voice recognition unit **130** and receives a voice input from the user.

(60) For example, the program playback function performs setting and change of a channel, volume, and the like of a television program or an Internet distributed program, playback of a recorded program, and the like in accordance with an input operation by the user of the input device **103** or the local terminal, and a voice input operation (hereinafter, it is simply referred to as a user's input operation, but when it is necessary to distinguish, it is referred to as a voice input operation) by the user recognized by the voice recognition unit **130**. The electronic program guide display function changes display, a channel, a time zone, and the like of the electronic program guide according to an input operation by the user.

(61) Furthermore, the recording setting function sets the recording date and time and the channel according to the input operation by the user. Data of the recorded program is stored in the memory

105 of the information processing device **100**. The external input switching function switches the external input according to the input operation by the user. The photograph regeneration function regenerates a picture captured on the designated date input according to the input operation by the user. The Internet browsing function executes the Internet search using the keyword input according to the input operation by the user.

(62) Furthermore, the image recognition function recognizes an input image captured by the camera **101** and input to the application unit **150**. In the image recognition, for example, an image feature amount extracted from the input image is collated with an image feature amount stored in advance by the feature DB **140** of a predetermined portion (for example, a face, a hand, or the like) of the user's body, thereby recognizing the predetermined portion. For example, the image recognition function identifies the user by face recognition, receives gesture input of the user, and recognizes a predetermined gesture from movement of a hand.

(63) Here, in a case where the history information and the preference information described above are stored for each of a plurality of users such as family members, it is necessary to identify the user in order to read out the history information and the preference information suitable for the user. Therefore, a user identifying function (profile function) for identifying a user is executed by the application unit **150**. The profile function identifies the user according to, for example, an input operation by the user or biometric information (for example, voice, face, or the like) of a human. As an example, one GUI is selected by a user's input operation on a graphic user interface (GUI) displayed for each user, and a user corresponding to the selected GUI is identified. Furthermore, for example, as described above, voice recognition by the voice recognition unit **130** or image recognition by the image recognition function is executed, and the user is identified.

(64) For example, in a case where a father says "show me a usual program", the user is identified as a father by voice recognition (for example, voiceprint matching) or image recognition (for example, face matching), and history information and preference information about the identified father are read from the history DB **172** and the preference DB **174**. Furthermore, it is also possible to identify the user with a keyword by voice. For example, in a case where a father says "show me the father's usual program", the father is identified by a keyword of "father", and history information and preference information about the identified father are read from the history DB **172** and the preference DB **174**.

(65) The search control unit **162** performs a process of searching for the content desired by the user based on the content selection information including the history information and the preference information according to the input operation by the user. For example, the search control unit **162** searches for a channel (a channel number or a channel name) or a program name currently being broadcast or distributed via the Internet, a channel or a program name of a recorded program, or the like based on content selection information including history information and preference information according to an input operation by the user. Then, the search control unit **162** determines the content obtained by the search as the selection candidate content, and generates content selection candidate information about the determined selection candidate content. Note that the recorded program may be stored in the memory **105** or may be stored in an external device such as a recorder connected via the communication I/F **104**.

(66) The selection control unit **164** performs a process of selecting one content (display content) to be displayed on the entire screen of the display **108** from the selection candidate contents (content selection candidate information) which is the search result by the search control unit **162**. In a case where the number of the selection candidate contents is one, the selection control unit **164** may select the one content. However, in a case where the number of the selection candidate contents is plural, the selection control unit selects one display content according to a predetermined criterion in order to select the content that is highly likely to be requested by the user.

(67) The predetermined criterion is, for example, priority based on content selection information including history information and preference information. The selection control unit **164**

determines the priority of each of the plurality of selection candidate contents according to one or both of the history information and the preference information of the content selection information. Note that the user is identified by the user's input operation, voice recognition, image recognition (face recognition), or the like as described above, and the history information and the preference information corresponding to the identified user are obtained from the history DB **172** and the preference DB **174**.

(68) For example, the selection control unit **164** increases the priority of a channel, a program, a genre, or the like frequently viewed by the viewing user or increases the priority of a channel, a program, a genre, or the like frequently recorded by the viewing user based on one or both of history information and preference information about the identified user (viewing user). The genre is a classification according to content contents, and examples of the genre include news, variety show, music, drama, movie, and documentary. As an example, the selection control unit **164** increases the priority of the channel/program/genre for each time zone or each day of the week in descending order of the viewing time or the recording time in the latest predetermined period (for example, one month). The selection control unit **164** determines the program with the highest priority as the display content.

(69) Note that the selection control unit **164** may calculate the priority according to a combination of users or the number of users. For example, based on the viewing history, it is possible to identify a channel/program/genre that is frequently viewed in a case where the family is together, a channel/program/genre that is frequently viewed in a case where only the parents are together, a channel/program/genre that is frequently viewed in a case where only the children are together, or the like, and to appropriately calculate the priority in each case. Furthermore, even in a case where the identification information about the user cannot be acquired, for example, the selection control unit **164** may increase the priority of a channel/program/genre that is frequently viewed, a program/genre that is frequently recorded, or the like, in the current time zone. Furthermore, the selection control unit **164** may increase the priority of, for example, the most recently selected channel, program, or genre based on only the viewing history regardless of the person or time. In addition, in a case where a recorded program is included in selection candidate content that is a search result, the selection control unit **164** may increase the priority of an unviewed program or decrease the priority of a viewed program.

(70) The display control unit **166** controls display of an image via the display **108**. For example, the display control unit **166** generates, according to a search result by the search control unit **162** and a selection result by the selection control unit **164**, an image according to the search result and the selection result to output and displays the image on the display **108**. In addition, the display control unit **166** causes the display **108** to display the application image input from the application unit **150**.

(71) For example, the display control unit **166** performs a process of displaying the image G1 of the selected display content on the entire screen of the display **108** according to the selection result by the selection control unit **164** (see FIG. 1). In addition, the display control unit **166** performs a process of generating the TV menu G2 corresponding to the selection candidate content and displaying the TV menu G2 on the screen of the display **108** according to the selection candidate content (content selection candidate information) which is the search result obtained by the search control unit **162** (see FIG. 1). At this time, the TV menu G2 is superimposed on the image G1 of the display content displayed on the entire screen of the display **108**, and is displayed in a predetermined region (for example, the lower region of the screen) of part of the screen of the display **108** (see FIG. 1). The TV menu G2 is automatically deleted, for example, in a case where there is no input operation by the user for a certain period of time after display. The display time of the TV menu G2 is set in advance, but may be changed based on user setting information, preference information, or the like.

(72) Note that, in the TV menu G2 described above, the display content varies depending on filter

selection according to the input operation by the user, but a display screen at the time of selecting a usual filter, a display image at the time of selecting a genre filter, and the like will be described later in detail. Note that the filter is a filter for selecting a channel or a program that the user wants to view, and various filters exist in the TV menu G2.

(73) The control unit **160** controls (processes) the entire information processing device **100** in addition to the search control unit **162**, the selection control unit **164**, and the display control unit **166**. For example, the control unit **160** performs a process of turning on the power of the entire information processing device **100** including the display **108** according to the input operation by the user (return from the standby state). In addition, the control unit **160** performs a process of activating the program playback application by the application unit **150** and displaying the image G1 of the display content selected by the selection control unit **164** on the display **108**. At this time, the control unit **160** also performs a process of superimposing and displaying the TV menu G2 on the image G1 of the display content.

(74) The functional configuration of the information processing device **100** according to the first embodiment has been specifically described above. Note that the functional configuration of the information processing device **100** illustrated in FIG. 3 is an example, and the present invention is not limited to this. For example, at least part of the functional configuration of the information processing device **100** illustrated in FIG. 3 may be in an external device (for example, a local terminal or a server on a cloud) communicatively connected to the information processing device **100**.

1-3. Display Screen at the Time of Selecting Usual Filter: TV Menu, Usual Program Guide

(75) Next, the first display screen (TV menu: first display image) at the time of selecting a usual filter according to the first embodiment will be described with reference to FIG. 4, and the second display screen (usual program guide: third display image) at the time of selecting a usual filter according to the first embodiment will be described with reference to FIG. 5. FIG. 4 is a diagram for describing the first display screen at the time of selecting a usual filter according to the first embodiment. FIG. 5 is a diagram for describing the second display screen at the time of selecting a usual filter according to the first embodiment.

(76) As illustrated in FIG. 4, for example, the image G1 of the display content is displayed on the entire screen of the display **108**, and the TV menu G2 is displayed in the lower region of the screen of the display **108**. The TV menu G2 is a user interface image (UI image) having an image G2a indicating the first selection column and an image G2b indicating the second selection column. The image G2a indicating the first selection column is an image in which item images corresponding to selection candidate contents of respective channels are disposed and indicated in a selectable manner in the horizontal direction of the screen of the display **108**. In addition, the image G2b indicating the second selection column is an image in which filter names (category information) of respective types are disposed and indicated in a selectable manner in the horizontal direction of the screen of the display **108**. In FIG. 4, for example, “soccer”, “usual program”, “BS”, “CS”, “advanced BS”, “advanced CS”, and the like are illustrated as filter names.

(77) Note that examples of the filter include filters of a terrestrial wave, a BS, and the like, filters (genre filters) of sports, a movie, and the like, filters of sports types such as soccer, baseball, and the like, but the present invention is not limited to this, and for example, may include categories such as a team filter, an actor filter, an original author filter, and the like. Furthermore, there may be a category of an external device filter such as an STB or a recorder, an external input filter such as HDMI (registered trademark), or an application filter such as a video on demand (VOD) application or a streaming application.

(78) Here, for example, in the TV menu G2 described above, when any of the item images is selected by the user's input operation, the image G1 of the display content is switched to the content image of the channel based on the selected item image. In addition, when any of the filters is selected, the image G1 of the display content may be switched to a content image corresponding

to the selected filter, and the TV menu G2 corresponding to the selected filter may be displayed. (79) For example, when a usual filter (filter name in FIG. 4: “usual program”) is selected, the display content and the selection candidate content (content selection candidate information) are determined based on the history information included in the content selection information. Then, as illustrated in FIG. 4, the image G1 of the determined display content is displayed on the entire screen of the display 108, and further, the TV menu G2 having item images corresponding to the determined selection candidate content and respective filter names are displayed in the lower region of the screen of the display 108. The respective item images are disposed in the screen of the display 108 in the horizontal direction in order of priority in the TV menu G2. For example, the respective item images are disposed in descending order of priority from the left. Further, when the usual filter is selected, only the item image in the TV menu G2 may be changed, and the image G1 of the display content may not be changed. Furthermore, for example, in a state where the selection frame or the cursor is placed on each filter of the TV menu G2, only the item images in the TV menu G2 are changed, and when the filter is selected by pressing a determination button provided in the remote controller, the display content may be changed to a display content corresponding to the usual filter or a display content of the content source corresponding to the filter.

(80) Note that the respective item images may be disposed in order of priority in the horizontal direction of the screen in the upper region of the screen of the display 108, or may be disposed in order of priority in the vertical direction of the screen in the left region or the right region of the screen of the display 108. Alternatively, the respective item images may be randomly displayed on the entire screen of the display 108 so that the priority can be recognized, and a display style thereof is not particularly limited.

(81) In addition, in a state where a usual filter (filter name in FIG. 4: “usual program”) is selected, for example, when a predetermined input operation (as an example, a pressing operation of a down button provided in the remote controller) is received, a usual program guide G3 is displayed on the entire screen of the display 108 as illustrated in FIG. 5. the usual program guide G3 is a user interface image having a program guide image G3a and an image G3b indicating the selection column. Program guide image G3a is an image in which item images corresponding to the selection candidate contents determined as described above are disposed and indicated for respective time zones. The respective item images are disposed in the screen of the display 108 in the longitudinal direction in order of priority in the program guide image G3a. For example, the respective item images are disposed in descending order of priority. In addition, the image G3b indicating the selection column is an image in which filter names (category information) of respective types are disposed and indicated in a selectable manner in the horizontal direction of the screen of the display 108. The image G3b indicating the selection column is displayed in the upper region of the screen of the display 108. In FIG. 5, as in FIG. 4, for example, “soccer”, “usual program”, “BS”, “CS”, “advanced BS”, “advanced CS”, and the like are illustrated as filter names. Here, an image indicating an operation corresponding to the process of switching between the TV menu G2 and the usual program guide G3 may be displayed. Specifically, for example, it is conceivable to display an image indicating a remote controller operation (for example, a down button pressing operation of the remote controller) or a voice utterance (for example, “open a program guide”) for switching to a screen including the usual program guide G3 on a screen including the TV menu G2 illustrated in FIG. 4.

(82) Here, the TV menu G2 and the usual program guide G3 has the usual filter (filter name in FIG. 4: “usual program”) at positions corresponding to each other. The corresponding positions in the TV menu G2 and the usual program guide G3 represent, for example, the same positional relationship in the column direction in which the filter names are disposed. In addition, the corresponding positions represent, for example, a positional relationship in which in a case where the TV menu G2 is displayed in the lower region of the screen of the display 108, the image G3b of the usual program guide G3 is displayed in the upper region (the display region opposite to the

display region of the TV menu G2 on the screen of the display 108) of the screen of the display 108. The filters other than the usual filter (for example, “soccer”, “BS”, and the like of the images Gb2 and G3b) may be displayed at corresponding positions, or all filters may be displayed at corresponding positions in the TV menu G2 and the usual program guide G3. In addition, at the time of switching processing between the TV menu G2 and the usual program guide G3, the image G2b indicating the second selection column of the TV menu G2 and the image G3b indicating the selection column of the usual program guide G3 may be seamlessly switched. For example, image display may be performed such that the image G2b indicating the second selection column of the TV menu G2 moves to a position corresponding to the image G3b indicating the selection column of the usual program guide G3.

(83) Note that, in the TV menu G2 or the usual program guide G3, the item image may include the recorded program and the like in addition to the current broadcast program of the channel obtained by the search, and the current broadcast program and the recorded program may be simultaneously displayed. That is, the search control unit 162 also outputs a search result including the recorded program of the searched channel. Here, it is also possible that only the recorded program is included and the program currently being broadcast is not included as the search target. In addition, the recorded program to be searched for may be limited to an unviewed program.

(84) In addition, in the TV menu G2 or the usual program guide G3, the name of the channel may be displayed in a predetermined display mode in the item image, and in addition, the title of the broadcast program currently broadcasted in the channel, the channel number, the mark of the channel or the broadcast station, and the like may be displayed, that is, information related to the channel, the program, and the like may be displayed. Furthermore, in order to clearly indicate the item image indicating the currently selected channel, the selected item image may be highlight-displayed so as to be different from other item images, or an icon such as “select” may be added to the selected item image. Furthermore, in the respective item images, information indicating a viewing state of each content, for example, a bar indicating a previous playback position (viewed position) may be displayed, and a “NEW” mark or the like may be displayed for an unviewed content.

(85) In addition, in the TV menu G2 or the usual program guide G3, in the item image, channel information, a content name (a program title or the like), a content outline, or information indicating whether the content is a currently broadcast program, whether it is a recorded program, whether it is a VOD item, whether it is an Internet distributed program, or the like may be displayed, or a still image or a moving image thumbnail may be displayed. The information about the channel and the content, the still image and the moving image thumbnail may be acquired from a distribution source of each content, or may be acquired from a different source such as a program information transmission server. Furthermore, the moving image thumbnail of the content displayed in the item image may be acquired from a moving image distribution server, a memory, or the like and played back before the user selects the content. For example, the moving image thumbnail of the broadcast program is acquired from the VOD server, or the moving image in which the broadcast program is recorded is played back in the item image as the moving image thumbnail of the broadcast program. In the still image and the moving image thumbnail, an early stage portion of the content may be used, an any position in the content may be used, or a moving image dedicated to the thumbnail may be used. As described above, various types of information such as a still image and a moving image may be acquired from various external devices such as a server and a recorder.

1-4. Priority at the Time of Selecting Usual Filter

(86) Next, the priority at the time of selecting a usual filter based on the content selection information (history information) according to the first embodiment will be described with reference to FIGS. 6 and 7. FIG. 6 is a first diagram for describing priority at the time of selecting a usual filter according to the first embodiment. FIG. 7 is a second diagram for describing priority at

the time of selecting a usual filter according to the first embodiment.

(87) As illustrated in FIG. 6 (upper stage), in history information J1 about the user in a certain time zone of a certain day of the week, the “channels being used” by the user are A to J. Further, the “channels and programs customarily watched at the time” by the user are “A” and “aaa”, and “D” and “ddd”. The “genre customarily watched by the user at the time” includes “news” and “variety show”. Upper alphabet letters represents channel names, and lower alphabet letters represents program names. In FIG. 6, the “channels being used” and the “channels and programs customarily watched at the time” are disposed in descending order of priority from the top in FIG. 6. The “genres customarily watched at the time” are disposed in descending order of priority from the left in FIG. 6. In addition, the priority is higher in the order of the “channel being used”, the “genres customarily watched at the time”, and the “channels and program customarily watched at the time”. The history information J1 as illustrated in FIG. 6 (upper stage) is stored in the history DB 172 for each day of the week and each time zone. Therefore, history information for each day of the week and each time zone is obtained, and history information for one week is stored in the history DB 172.

(88) Based on such history information J1, the priority of the channel and the program (content) is determined by the selection control unit 164. For example, when information J2 related to channels and programs as illustrated in FIG. 6 (middle stage) is obtained from a program guide on the Internet, an electronic program guide from each broadcasting station, or the like, the channels and programs are sorted in order of priority as illustrated in FIG. 6 (lower stage) based on the history information J1. For example, the channels and programs are sorted in descending order of priority from the top in FIG. 6. Here, if the channel name is “C” and the program name is “ccc”, it is the “channel and program customarily watched at the time”, and thus the priority is increased.

(89) On the other hand, when information J3 related to channels and programs as illustrated in FIG. 7 (upper stage) is obtained from a program guide on the Internet, an electronic program guide from each broadcasting station, or the like, the channels and programs are sorted in order of priority as illustrated in FIG. 7 (lower stage) based on the history information J1. For example, the channels and programs are sorted in descending order of priority from the top in FIG. 7. Here, if the channel name is “C” and the program name is “xxx”, although it is the channel “C” that is customarily watched, the program name is different, so that it is not the “channel and program that are customarily watched at the time”, and thus the priority is lowered.

(90) As described above, since matching of both the channel and the program is confirmed in determination of the priority at the time of selecting a usual filter, it is possible to suppress an offer of a program that the user is not interested in, that is, the user does not want. That is, in a case where matching of only channels is confirmed, even when the program is changed to a program not desired by the user, a program not desired by the user may be provided to the user. However, as described above, by confirming matching of both the channel and the program, in a case where the program is changed to a program not desired by the user, the program does not match even if the channel match. Therefore, it is possible to suppress an offer of the program not desired by the user to the user.

(91) Examples of the genre of programs include news, variety show, sports, documentary, drama, movie, and music. By setting a preferred genre in advance, the user may search for a more detailed genre, for example, search for a content having a plurality of genre attributes like a documentary related to music. Furthermore, although the genre of the program is described here as an example, the genre of the content is not limited to this example, and may be, for example, moving image, music, image, application, or the like, or may be a content genre other than the moving image content such as Jazz, Rock, or Ballard.

(92) Note that the priority of the selection candidate content may be determined based on the preference information in addition to the history information J1 of the user. Furthermore, the priority may be determined in consideration of settings (user setting information) made by the user

for the channel or the content. For example, processing in which the user can designate the priority itself of the channel or the content, or the priority of the program included in the viewing schedule list, the content added to the favorite, or the program whose recording has been reserved is increased may be performed. Furthermore, the priority may be determined by combining the preference information or history information about the user with the preference information or history information about another person such as a family member or a friend of the user, or the priority may be determined only from the preference information or history information about another person such as a family member or a friend of the user.

1-5. First Display Process at the Time of Selecting Usual Filter: TV Menu

(93) Next, the first display process at the time of selecting a usual filter according to the first embodiment will be described with reference to FIG. 8. FIG. 8 is a flowchart illustrating a flow of the first display process at the time of selecting a usual filter according to the first embodiment.

(94) As illustrated in FIG. 8, when the control unit **160** of the information processing device **100** detects an input operation by the user for turning on the power of the information processing device **100** (step S1), the control unit **160** turns on the power of the information processing device **100** (step S2). As a result, the display **108** turns on a light.

(95) The control unit **160** determines whether the power is turned on at intervals of more than a predetermined time (step S3). When it is determined that the power is not turned on at intervals of more than the predetermined time (NO in step S3), the display control unit **166** performs a process of displaying the display content of the channel selected when the power is turned off on the display **108** (step S4), and ends the process.

(96) On the other hand, when the power is turned on at intervals of more than the predetermined time (YES in step S3), the search control unit **162** searches for and determines a selection candidate content from a plurality of contents corresponding to the current time zone based on the content selection information (step S5). The selection control unit **164** selects and determines one display content (content to be displayed on the entire screen of the display **108**) from among the selection candidate contents obtained by the search by the search control unit **162** based on the content selection information (step S6).

(97) The display control unit **166** performs a process (channel selection) of displaying the image G1 (video) of the display content determined by the selection control unit **164** on the display **108** and a process of generating the TV menu G2 corresponding to one or more selection candidate contents and displaying the TV menu G2 on the display **108** (step S7). For example, based on the content selection information, the selection control unit **164** increases the priority of the content that is frequently viewed in the current time zone and the day of the week, and sets the content with the highest priority as the display content. In addition, the display control unit **166** generates the TV menu G2 by disposing the item images corresponding to respective selection candidate contents in order of priority.

(98) Next, the selection control unit **164** determines whether another channel has been selected by the user's input operation (step S8). When another channel is selected by the user's input operation (YES in step S8), the selection control unit **164** determines the selected another channel as the display channel, and the display control unit **166** performs a process of displaying the image (video) G1 of the display content of the determined display channel on the display **108** (step S9). Thereafter, when a certain period of time (for example, about several seconds to about several tens of seconds) has elapsed from the display of the TV menu G2 or the selection of another channel and a timeout occurs (YES in step S10), the display control unit **166** performs a process for ending the display of the TV menu G2 (step S11).

(99) Note that the final channel selection result together with time information and identification information about the user is stored in the history DB **172** as a viewing history of the user, and is used when the priority is calculated by the selection control unit **164** at the time of the next channel selection. By repeating such processing, the information about the viewing history of the user is

accumulated, and the selection candidate content is obtained based on the history information in which the information amount is increased. As a result, since the possibility that the selection candidate content is a content desired by the user is improved, the user can view the desired content more reliably.

(100) According to the first display process as described above, in the determination of the content selection candidate information, the content is searched for based on the content selection information (for example, history information) of the user, and the content obtained by the search is determined as the selection candidate content. In the determination of the display content, the priority of each selection candidate content is determined based on the content selection information about the user, and the channel having the highest priority is determined as the display content. In addition, respective selection candidate contents are disposed in order of priority, the TV menu G2 including item images corresponding to respective selection candidate contents disposed in order of priority and filter names is created, and the TV menu G2 is displayed together with the image G1 of the display content.

(101) In this way, since the content that is highly likely to be desired by the user is automatically presented to the user, the user can view the desired content without executing a complicated operation of selecting the channel. As a result, the user can easily view the desired content. In a case where the display content is not the content desired by the user, the user immediately selects the channel again. Even in this case, since the TV menu G2 corresponding to the selection candidate contents is displayed, the user can visually recognize the TV menu G2 and change the content (channel change). As a result, the user can easily select and view the desired content.

(102) Note that the usual filter is preferentially used at the time of activation, but the present invention is not limited to this, and the usual filter may not be preferentially used, or the usual filter may be preferentially used according to user selection (user setting information). For example, the control unit **160** determines whether the usual filter of the TV menu G2 has been used a specified number of times (for example, 10 times) or more, and whether the function introduction of the usual filter has never been displayed. In a case where the usual filter of the TV menu G2 is used a specified number of times (for example, 10 times) or more, or in a case where the function introduction of the usual filter is never displayed, the control unit **160** presents, to the user, a user interface image in which it is possible to select whether to preferentially use the usual filter. Then, in a case where the user selects to preferentially use the usual filter according to the user selection in the user interface image, the control unit **160** performs a process of preferentially using the usual filter.

(103) Furthermore, in a case where the searched contents are a plurality of contents having an order such as a series of drama or a movie series (in a case where a plurality of contents having the same title name is searched for), for example, the control unit **160** may perform a process of listing contents in ascending order of numbers (story number, series number, and the like), selecting (an example of priority) the content having the lowest number except for contents that can be determined to have been viewed or almost watched up to before the staff roll, and playing back the content on the display **108**. By including not only the unviewed content but also the content of which the viewing is interrupted in the middle as the contents to be selected, it is possible to play back the content from the beginning or the subsequent content without excluding the content that has been watched halfway. Furthermore, the information processing device **100** may preferentially select content that has been watched halfway. In this case, in step S5, it is possible to search for and determine the selection candidate content from contents other than the content corresponding to the current time zone.

(104) Furthermore, in a case where there is a plurality of serial drama series, the content with the highest priority may be selected, and then respective series of drama may be displayed in the item images. In this case, in a case where the user designates another serial drama series, processing in which the content having the highest priority among the another serial drama series is selected and

played back, and then other episodes in the series are displayed as item images may be performed. Furthermore, in a case where a viewing user can be identified by voice recognition or the like, the information processing device **100** can implement personalized playback control by referring to the viewing history of the viewing user and referring to the viewing state of the viewing user regarding the searched content.

(105) As described above, the information processing device **100** searches for content, and in a case where a plurality of contents has been searched for, automatically selects the image **G1** of the display content having the highest possibility of being requested by the user according to a predetermined criterion and displays the image **G1** on the display screen, and displays the TV menu **G2** corresponding to the selection candidate contents as the search result. As a result, even in a case where there is a plurality of selection candidates, the content is automatically selected and displayed, so that the user's trouble is saved and the user can easily view the desired content. In addition, by displaying the TV menu **G2** corresponding to the selection candidate contents, it is possible to support the user selecting another content, so that the user can easily select and view the desired content.

1-6. Second Display Process at the Time of Selecting Usual Filter: Usual Program Guide

(106) Next, the second display process at the time of selecting a usual filter according to the first embodiment will be described with reference to FIG. **9**. FIG. **9** is a flowchart illustrating a flow of the second display process at the time of selecting a usual filter according to the first embodiment.

(107) As illustrated in FIG. **9**, the control unit **160** of the information processing device **100** determines whether the TV menu **G2** at the time of selecting a usual filter is displayed (step **S21**). When the TV menu **G2** at the time of selecting a usual filter is displayed on the display **108** (YES in step **S21**), it is determined whether a first input operation has been performed (step **S22**). An example of the first input operation includes a pressing operation of a down button provided in the remote controller.

(108) In a case where the first input operation has not been performed (NO in step **S22**), a certain period of time (for example, about several seconds to several tens of seconds) has elapsed from display of the TV menu at the time of selecting a usual filter, and a timeout has occurred (YES in step **S23**), the display control unit **166** performs control to end display of the TV menu **G2** (step **S24**). On the other hand, when the first input operation is performed (YES in step **S22**), the display control unit **166** ends the display of the TV menu **G2**, and performs control to output the usual program guide **G3** to display it on the display **108** (step **S25**).

(109) Next, the control unit **160** determines whether another channel has been selected by the user's input operation (step **S26**). When another channel is selected by the user's input operation (YES in step **S26**), the selection control unit **164** selects the selected another channel as the display channel, and the display control unit **166** performs control to display the image (video) **G1** of the display content of the selected display channel on the display **108** (step **S27**), and ends the display of the usual program guide **G3** (step **28**).

(110) On the other hand, when another channel is not selected by the user's input operation (NO in step **S26**), the control unit **160** determines whether there is a second input operation (step **S29**). When the second input operation is performed (YES in step **S29**), display control unit **166** ends the display of the usual program guide **G3**, and performs a process of displaying the original TV menu **G2** on the display **108** (step **S30**). Thereafter, the process returns to step **S21**. An example of the second input operation includes a pressing operation of an up button provided in the remote controller.

(111) According to the second display process, it is possible to present the user with the selection candidate content that is highly likely to be desired by the user with the usual program guide **G3**, and thus, the user can avoid a complicated operation of selecting a channel and easily select and view the desired content. In addition, since it is possible to smoothly switch between the usual program guide **G3** and the TV menu **G2** at the time of selecting a usual filter, it is possible to

improve user convenience. In addition, in steps S21 to S30 of the flowchart illustrated in FIG. 9, processing of switching between the usual program guide G3 and the TV menu G2 at the time of selecting a usual filter is described. When the first input operation is performed even in a case where a filter other than usual filter is selected in the TV menu G2, the display may be switched to the program guide on the entire screen corresponding to the filter other than the usual filter selected in the TV menu G2.

1-7. Third Display Process at the Time of Selecting Usual Filter: Voice Input Operation

(112) Next, the third display process (voice input operation) at the time of selecting a usual filter according to the first embodiment will be described with reference to FIG. 8. FIG. 8 is a flowchart illustrating a flow of the first display process at the time of selecting a usual filter according to the first embodiment.

(113) In step S1 illustrated in FIG. 8, the input operation is an input operation by voice (voice input operation). In this voice input operation, when the information processing device 100 detects the user's utterance voice (for example, the “predetermined call”+“show me the usual program”) using the microphone 102, the voice recognition unit 130 recognizes the voice input. The power supply state of the information processing device 100 at this time is a standby state in which a voice can be detected by the microphone 102 (that is, a state in which power is being supplied to the processor 112 and the microphone 102), and the display 108 is turned off (that is, a state in which power supply to the display 108 is not performed). After voice input recognition in step S1, processing similar to that in FIG. 8 is performed in and after step S2 of turning on the power.

(114) For example, when the user utters the “predetermined call”+“show me the usual program” in step S1, the power is turned on in step S2, and the image G1 of the display content and the TV menu G2 corresponding to the selection candidate contents are displayed in step S7 after steps S3 to S6 (see FIG. 4). The user can activate the device from the standby state by the voice input, and the image G1 of the desired display content and the TV menu G2 corresponding to the selection candidate contents are automatically displayed on the screen. As a result, the user can more easily activate the information processing device 100 by the voice input operation, avoid a complicated operation of selecting a channel, and more easily view the desired content. Note that, in step S1, in a case where the user utters a request for a program to be regularly viewed, such as the “predetermined call”+“show me the usual program”, after the power is turned on in step S2, the process may proceed to the content determination process in step S4 without determining whether the power is turned on after the predetermined elapsed time or more in step S3. In this case, in a case where the user has uttered to turn on the power of the television, such as the “predetermined call”+“turn on the television” in step S1, the determination process in step S3 may be performed.

(115) Note that the reception of the utterance of the user may be started without a predetermined call by the user performing the input operation on a button or the like disposed in the remote controller or performing the input operation on a mobile terminal such as a smartphone.

Furthermore, the voice reception may be performed not by the information processing device 100 but by the remote controller or the smartphone. Furthermore, the voice input can be received by a voice device (for example, a smart speaker) connected to the information processing device 100. The voice device may transmit the detected user voice to a voice recognition server on the cloud. In this case, the information processing device 100 is automatically powered on by a control signal transmitted from a voice device or a server on the cloud, and displays the content requested by the user.

(116) Furthermore, after activation according to the utterance (or input operation) of the user, the control unit 160 may determine whether the setting for STB viewing is completed in a case where the STB is connected to the information processing device 100, and execute the viewing process by the STB in a case where the setting for STB viewing is completed, or execute the viewing process by the information processing device 100 (built-in tuner) in a case where the setting for STB viewing is not completed. That is, the processing can be changed according to the setting status of

the connected device.

(117) As described above, even in the standby state, the information processing device **100** can receive a voice input by the user, and can turn on the power and display the content requested by the user when the voice input (content output instruction) requesting the content playback is detected. Therefore, the user can more easily activate the information processing device **100** by the voice input operation, avoid complicated operation of selecting a channel, and more easily view the desired content.

1-8. Usual Filter Setting Screen

(118) Next, a usual filter setting screen (setting image) according to the first embodiment will be described with reference to FIGS. **10** and **11**. FIG. **10** is a first diagram for describing the usual filter setting screen according to the first embodiment. FIG. **11** is a second diagram for describing the usual filter setting screen according to the first embodiment.

(119) As illustrated in FIG. **10**, the display control unit **166** performs a process of displaying a setting image **G4** for checking whether to preferentially select a usual program in a partial region of the display **108**. The control unit **160** sets whether to preferentially select a usual program in accordance with an input operation by the user. When the user performs an input operation of selecting “Yes”, the user setting information includes information indicating that the usual program is preferentially selected, and when the user performs an input operation of selecting “No”, the user setting information includes information indicating that the usual program is not preferentially selected. In a case where the user setting information includes information indicating that the usual program is preferentially selected, the control unit **160** performs a process of preferentially selecting the usual program. In this way, since the user can set whether to preferentially select a usual program, a degree of freedom of selection by the user is increased, and convenience of the user can be improved.

(120) Furthermore, as illustrated in FIG. **11**, the display control unit **166** performs a process of displaying a setting image **G5** related to the usual channel list (usual filter) on the display **108**. The setting image **G5** is a user interface image (UI image), and when the user performs an input operation on the setting image **G5**, it is possible to set whether to preferentially select a usual program, a broadcast, a favorite channel list, a genre, a user input keyword or genre, and the like. Since the user can perform an input operation of selecting a desired element in each item and perform setting for each item, the degree of freedom of selection by the user is increased, and convenience of the user can be improved.

(121) However, in the case of continuous TV viewing use (display within a predetermined time from previous activation), the display control unit **166** may restrict preferentially selecting a channel of a usual program, that is, the use of the usual filter. For example, it is possible to prevent the channel from changing when the user temporarily turns off the television and comes back, such as when the user use a rest room for a short time, so that the convenience of the user can be improved. In this case, for example, the setting image **G5** may include an item that allows the user to set a predetermined time for which channel selection based on a usual filter is restricted.

(122) In addition, in a case where a favorite filter (filter name: “favorite”) has been last selected in the TV menu **G2**, for example, in a case where the favorite filter has been selected when the power is turned off, the display control unit **166** may restrict preferentially selecting a channel of a usual program, that is, the use of the usual filter. As a result, the favorite content can preferentially be displayed, so that the convenience of the user can be improved.

1-9. Summary of First Embodiment

(123) As described above, according to the first embodiment, the information processing device **100** acquires the content selection information (for example, history information such as a viewing history and a recording reservation of the user, and preference information about the user) about the user, and acquires the content output instruction (for example, an instruction by a user's input operation or voice input operation) related to the content selection information. When acquiring the

content output instruction, the information processing device **100** determines the display content and selection candidate information (selection candidate content) about the content based on the content selection information to output image **G1** of the display content and the TV menu **G2**, which is an example of the first display image corresponding to the selection candidate information about the content to the display **108**. As a result, it is possible to automatically present the user with the content that is highly likely to be desired by the user, so that the user can easily view the desired content. Furthermore, since it is possible to present the user with the selection candidate content that is highly likely to be desired by the user, the user can easily select and view the desired content.

(124) Furthermore, the TV menu **G2** at the time of selecting a usual filter is created based on one or both of the service (channel) and the content used by the user within a predetermined period (for example, one month). As a result, it is possible to reliably provide the content that is highly likely to be desired by the user according to one or both of the service and the content used by the user within the predetermined period, so that the user can easily select the desired content. As a result, the user can easily view the desired content.

(125) In addition, the TV menu **G2** at the time of selecting a usual filter changes according to one or both of the time zone and the day of the week. As a result, it is possible to provide the content that is highly likely to be desired by the user according to one or both of the time zone and the day of the week, so that the user can easily select the desired content. As a result, the user can easily view the desired content.

(126) In addition, the TV menu **G2** at the time of selecting a usual filter includes a plurality of pieces of selectable category information (for example, various filters) corresponding to one or both of the content acquisition means (for example, terrestrial broadcasting, satellite broadcasting, set top box (STB), HDMI (registered trademark), or the like) and the content type (for example, sports, drama, or the like). As a result, since the user can select category information, it is possible to perform a process corresponding to the category information selected by the user, and the degree of freedom of selection by the user can be increased to improve convenience of the user.

(127) In addition, the TV menu **G2** at the time of selecting a usual filter includes lower category information (for example, sport name, drama name, or the like) related to the content type (for example, sports, drama, or the like). In addition, the TV menu **G2** at the time of selecting a usual filter includes lower category information (for example, team name or actor name) related to an element (for example, team or actor) included in the content. As a result, since the lower category information related to the type or element of the content is provided to the user, the user can grasp the lower category information related to the type or element of the content and easily select the desired content. As a result, the user can easily view the desired content.

(128) In addition, in the TV menu **G2** at the time of selecting a usual filter, content selection candidates (selection candidate contents) are disposed in an order based on the content selection information. As a result, the content selection candidates are disposed in an order suitable for the user and provided to the user, so that the user can easily select the desired content. As a result, the user can easily view the desired content.

(129) In addition, the control unit **160** performs a process of switching from an output of the display content and the TV menu **G2** at the time of selecting a usual filter to an output for displaying the usual program guide **G3**, which is an example of the third display image corresponding to the content information interface, on the entire screen based on the user input operation for the TV menu **G2** at the time of selecting the usual filter. This makes it possible to provide a content selection candidate suitable for the user with the usual program guide **G3**, so that the user can easily select the desired content. As a result, the user can easily view the desired content.

(130) In addition, the TV menu **G2** at the time of selecting a usual filter and the usual program guide **G3** include selectable category information (for example, usual filter “usual program”) at

corresponding positions. As a result, even when the TV menu G2 at the time of selecting a usual filter and the usual program guide G3 are switched, the category information is easily visually recognized, and the switching can be performed without discomfort, so that the convenience of the user can be improved.

(131) In addition, in the usual program guide G3, content selection candidates (selection candidate contents) are disposed in an order based on the content selection information. As a result, the content selection candidates are disposed in an order suitable for the user and provided to the user, so that the user can easily select the desired content. As a result, the user can easily view the desired content.

(132) Furthermore, the control unit 160 performs a process of outputting the setting images G4 and G5, which are examples of the setting information for selecting to display the image G1 of the display content selected based on the content selection information at the time of activation, and a process of displaying the image G1 of the display content selected based on the content selection information at the time of activation based on user selection for the setting images G4 and G5. As a result, the user can select whether to display the image G1 of the display content selected based on the content selection information at the time of activation, and thus, it is possible to increase the degree of freedom of selection by the user and improve convenience of the user.

(133) Furthermore, the content output instruction may be an utterance of the user. As a result, the user can more easily output the image G1 of the display content and the TV menu G2 to the display 108 by the voice input operation, so that the user can more easily view the desired content.

(134) Furthermore, the control unit 160 performs a process of identifying the user based on the utterance of the user, and in the process of determining the display content and the selection candidate information (selection candidate content) of the content, determines the display content and the selection candidate information about the content based on the content selection information about the identified user. As a result, since the content selection information suitable for the user is used, it is possible to more reliably present the selection candidate content that is highly likely to be desired by the user to the user. As a result, the user can easily select and view the desired content.

2. Second Embodiment

(135) The second embodiment is different from the first embodiment in a display screen at the time of selecting a genre filter and a process related to the display screen. Hereinafter, differences from the first embodiment will be mainly described.

2-1. Display Screen at the Time of Selecting Genre Filter: TV Menu, Preference Program Guide

(136) The first display screen (TV menu: second display image) at the time of selecting a genre filter according to the second embodiment will be described with reference to FIG. 12, and the second display screen (preference program guide: third display image) at the time of selecting a genre filter will be described with reference to FIG. 13. FIG. 12 is a diagram for describing the first display screen at the time of selecting a genre filter according to the second embodiment. FIG. 13 is a diagram for describing the second display screen at the time of selecting a genre filter according to the second embodiment.

(137) As illustrated in FIG. 12, for example, the image G1 of the display content is displayed on the entire screen of the display 108, and the TV menu G2 is displayed in the lower region of the screen of the display 108. The TV menu G2 is a user interface image (UI image) having an image G2a indicating the first selection column and an image G2b indicating the second selection column. The image G2a indicating the first selection column is an image in which item images corresponding to selection candidate contents of respective channels are disposed and indicated in a selectable manner in the horizontal direction of the screen of the display 108. The image G2a also includes, for example, an emblem image of a team corresponding to the selection candidate content. The image G2b indicating the second selection column is an image in which filter names (category information) of respective types are disposed and indicated in a selectable manner in the horizontal

direction of the screen of the display **108**. In FIG. **12**, for example, “soccer”, “terrestrial digital”, “sports”, “CS”, “advanced BS”, “advanced CS”, and the like are illustrated as filter names.

(138) Here, as in the first embodiment, for example, in the TV menu **G2** described above, when any of the item images is selected by the user's input operation, the image **G1** of the display content is switched to the content image of the channel based on the selected item image. When any of the filters is selected, the image **G1** of the display content is switched to the content image corresponding to the selected filter, and the TV menu **G2** corresponding to the selected filter is displayed.

(139) For example, in a case where a sports filter (filter name in FIG. **12**: “sports”) which is a genre filter is selected, the display content and the selection candidate content (content selection candidate information) are determined based on the preference information included in the content selection information. Then, as illustrated in FIG. **12**, the image **G1** of the determined display content is displayed on the entire screen of the display **108**, and further, the TV menu **G2** having item images corresponding to the determined selection candidate contents and respective filter names are displayed in the lower region of the screen of the display **108**. The respective item images are disposed in the screen of the display **108** in the horizontal direction in order of priority in the TV menu **G2**. For example, the respective item images are disposed in descending order of priority from the left. Further, the emblem image of the team is also displayed at the position corresponding to the selection candidate content. For example, in addition to the illustration corresponding to the selection candidate information such as the emblem image of the team, related information such as an image or an explanatory sentence corresponding to the selection candidate information such as a picture of a player belonging to the team or an explanatory sentence of the team may be displayed. The related information may be displayed so as to indicate that the related information is commonly related to the plurality of pieces of selection candidate information.

(140) In addition, in a state where the sports filter (filter name in FIG. **12**: “sports”) is selected, for example, when a predetermined input operation (as an example, a pressing operation of a down button provided in the remote controller) is received, a preference program guide **G6** is displayed on the entire screen of the display **108** as illustrated in FIG. **13**. The preference program guide **G6** is a user interface image having a program guide image **G6a** and an image **G6b** indicating the selection column. Program guide image **G6a** is an image in which item images corresponding to the selection candidate contents determined as described above are disposed. The respective item images are disposed in the screen of the display **108** in the longitudinal direction in order of priority in the program guide image **G6a**. For example, the respective item images are disposed in descending order of priority. In addition, the image **G6b** indicating the selection column is an image in which filter names (category information) of respective types are disposed and indicated in a selectable manner in the horizontal direction of the screen of the display **108**. The image **G6b** indicating the selection column is displayed in the upper region of the screen of the display **108**. In FIG. **13**, as in FIG. **12**, for example, “soccer”, “terrestrial digital”, “sports”, “CS”, “advanced BS”, “advanced CS”, and the like are illustrated as filter names.

(141) Here, the TV menu **G2** and the preference program guide **G6** have the sports filter (filter name in FIG. **12**: “sports”) at positions corresponding to each other. In the TV menu **G2** and preference program guide **G6**, the corresponding positions represent, for example, the same positional relationship in the column direction in which the filter names are disposed. Furthermore, the corresponding positions represent, for example, a positional relationship in which in a case where the TV menu **G2** is displayed in the lower region of the screen of the display **108**, the image **G6b** of the preference program guide **G6** is displayed in the upper region (the display region opposite to the display region of the TV menu **G2** on the screen of the display **108**) of the screen of the display **108**.

(142) Note that, it is possible to apply a display mode similar to that of the TV menu **G2** or the respective item images of the usual program guide **G3**, that is, the display mode according to the

first embodiment, to the respective item images of the preference program guide **G6**, and for example, channel information, a content name (program title or the like), a content outline, or information indicating whether the content is a current broadcast program, whether it is a recorded program, whether it is a VOD item, whether it is an Internet distributed program, or the like may be displayed, or a still image or a moving image thumbnail may be displayed.

(143) Furthermore, the genre filter (see FIGS. **12** and **13**) may be included in the selection column including the usual filter (see FIGS. **4** and **5**) in the TV menu **G2** or the usual program guide **G3**. That is, the TV menu **G2** or the usual program guide **G3** may include the genre filter in addition to the usual filter in the selection column. For example, the TV menu **G2** (the TV menu **G2** in which the usual filter is selected) determined based on the content selection information is displayed at the time of activation, and the TV menu **G2** includes the genre filter.

2-2. Priority at the Time of Selecting Genre Filter

(144) Next, priority at the time of selecting a genre filter according to the second embodiment will be described with reference to FIGS. **14A**, **14B**, **14C**, **14D**, and **15**. FIGS. **14A**, **14B**, **14C**, and **14D** are first diagrams for describing priority at the time of selecting a genre filter according to the second embodiment. FIG. **15** is a second diagram for describing priority at the time of selecting a genre filter according to the second embodiment.

(145) As illustrated in FIGS. **14A**, **14B**, **14C**, and **14D**, the selection control unit **164** changes the priority for determining the display order of the item images in the TV menu **G2** according to the information amount of the content selection information. In a case where the preference information is obtained from the history information included in the content selection information, it is determined that the longer the viewing time of the content is, the more the user likes the content, the information about the content is added to the preference information, and the preference information is updated. That is, the preference of the user is learned from the history information and the information is acquired, and the acquired information is reflected in the preference information about the user. Therefore, as the information amount of the history information increases, the information amount of the preference information also increases. Note that the preference information is not only obtained from the history information, but may be set by a user's input operation.

(146) In a case where the information amount of the preference information is insufficient, as illustrated in FIG. **14A**, the priority of the “viewable channel” is set higher than that of the “unviewable channel”. When the information amount of the preference information exceeds a first predetermined value, as illustrated in FIG. **14B**, the “favorite small genre” is set in the “viewable channel”, and the priority of the “favorite small genre” is set to be the highest in the “viewable channel”. When the information amount of the preference information exceeds a second predetermined value (>the first predetermined value), as illustrated in FIG. **14C**, the “favorite team” in addition to the “favorite small genre” is set in “viewable channel”, and the priority of the “favorite team” is set to be the highest “in the viewable channel”. Alternatively, as illustrated in FIG. **14D**, only “favorite team” is set in the “viewable channel”, and the priority of the “favorite team” is set to the highest “in the viewable channel”.

(147) When the information amount of the preference information exceeds the second predetermined value, for example, as illustrated in FIG. **15**, as preference information **J4** of the user, the “favorite sports team” of the user is set to “*** baseball team” and “*** soccer team”, and the “favorite small genre” of the user is set to “soccer”, “baseball”, and “golf”. The priority of the “favorite sports team” is higher than that of the “favorite small genre”. In addition, the “favorite sports team” and the “favorite small genre” are disposed in descending order of priority from the left in FIG. **15**. That is, the priority of the “*** baseball team” is higher than that of the “*** soccer team”, the priority of “soccer” is higher than that of “baseball” and that of “golf”, and the priority of “baseball” is higher than that of “golf”. The preference information **J4** of the user is stored in the preference DB **174** for each user. Note that there is sports as a category, there are baseball, soccer,

and golf as a lower category thereof, and there is team as a lower category thereof.

(148) The priority of the channel and the program (content) is determined by the selection control unit **164** based on such preference information **J4**. For example, channels and programs obtained from a program guide on the Internet, an electronic program guide from each broadcasting station, or the like are sorted in order of priority as illustrated in FIG. **15** (lower part) based on preference information **J4** as illustrated in FIG. **15** (upper part), and are sorted in descending order of priority from the left in FIG. **15**, for example. The priority of the “program that can be watched (for example, the program of the channel being used or the free program)” is higher than that of the “program that cannot be viewed”. In addition, in the “program that can be watched”, the priority is lowered in the order of the “**** baseball team program”, the “**** soccer team program”, the “soccer program”, the “baseball program”, the “golf program”, and “others”. Respective selection candidate contents are disposed based on such priority, and the TV menu **G2** having item images corresponding to the disposed selection candidate contents is generated.

(149) Note that the filter includes drama, variety show, movie, music, and the like in addition to sports as in the first embodiment described above. For example, in the drama, an original author name as a favorite original author or an actor name may be set, and the domestic drama may be set as a favorite small genre. An entertainer name may be set in the variety show, and a singer name may be set in the music.

2-3. First Display Process at the Time of Selecting Genre Filter: TV Menu

(150) Next, the display process performed by the information processing device **100** according to the second embodiment, that is, the first display process (TV menu) at the time of selecting a genre filter will be described with reference to FIG. **16**. FIG. **16** is a flowchart illustrating a flow of the first display process at the time of selecting a genre filter according to the second embodiment.

(151) As illustrated in FIG. **16**, the control unit **160** of the information processing device **100** determines whether a genre filter in the TV menu **G2** is selected (step **S31**). When the genre filter in the TV menu **G2** is selected (YES in step **S31**), the search control unit **162** searches for and determines the selection candidate content from a plurality of contents corresponding to the current time zone based on the genre related to the selected genre filter, that is, the selected genre and the content selection information (for example, preference information) (step **S32**). The selection control unit **164** selects and determines one display content (content to be displayed on the entire screen of the display **108**) from among the selection candidate contents obtained by the search by the search control unit **162** based on the selected genre and the content selection information (step **S33**).

(152) The display control unit **166** performs a process (channel selection) of displaying the image **G1** (video) of the display content selected by the selection control unit **164** on the display **108** and a process of generating the TV menu **G2** corresponding to one or more selection candidate contents and displaying the TV menu **G2** on the display **108** (step **S34**). For example, the selection control unit **164** sets the priority of a favorite sport, a sports team, or the like to be high based on the content selection information, and sets the content having the highest priority as the display content. In addition, the display control unit **166** generates the TV menu **G2** by disposing the item images corresponding to respective selection candidate contents in order of priority.

(153) Next, the selection control unit **164** determines whether another channel has been selected by the user's input operation (step **S35**). When another channel is selected by the user's input operation (YES in step **S35**), the selection control unit **164** determines the selected another channel as the display channel, and the display control unit **166** performs a process of displaying the image (video) **G1** of the display content of the determined display channel on the display **108** (step **S36**). Thereafter, when a certain period of time (for example, about several seconds to about several tens of seconds) has elapsed from the display of the TV menu **G2** or the selection of another channel and a timeout occurs (YES in step **S37**), the display control unit **166** performs a process for ending the display of the TV menu **G2** (step **S38**).

(154) Note that the final channel selection result together with time information and identification information about the user is stored in the history DB **172** as a viewing history of the user, and is also stored in the preference DB **174** as the preference information, and is used when the priority is calculated by the selection control unit **164** at the time of the next channel selection. By repeating such processing, the information about the viewing history and the preference of the user is accumulated, and the selection candidate content is obtained based on the history information and the preference information in which the information amount is increased. As a result, since the possibility that the selection candidate content is a content desired by the user is improved, the user can view the desired content more reliably.

(155) According to the first display process as described above, in the determination of the content selection candidate information, the content is searched for based on the selected genre and the content selection information (for example, preference information) of the user, and the content obtained by the search is determined as the selection candidate content. In the determination of the display content, the priority of each selection candidate content is determined based on the content selection information about the user, and the channel having the highest priority is determined as the display content. In addition, respective selection candidate contents are disposed in order of priority, the TV menu G2 including item images corresponding to respective selection candidate contents disposed in order of priority and filter names is created, and the TV menu G2 is displayed together with the image G1 of the display content.

(156) In this way, since the content that is highly likely to be desired by the user is automatically presented to the user, the user can view the desired content without executing a complicated operation of selecting the channel. As a result, the user can easily view the desired content. In a case where the display content is not the content desired by the user, the user immediately selects the channel again. Even in this case, since the TV menu G2 corresponding to the selection candidate contents is displayed, the user can visually recognize the TV menu G2 and change the content (channel change). As a result, the user can easily select and view the desired content.

(157) Note that the display content is automatically selected at the time of selecting a genre filter, and the image of the selected content is displayed on the display **108**, but the present invention is not limited to this, and the display content may not be selected, and the image of the content being displayed may be displayed as it is. That is, the process in step S33 and the display content display process in step S34 illustrated in FIG. **16** may not be performed. However, in a case where that the genre filter is automatically selected at the time of activation is set, the display content may be automatically selected at the time of selecting a genre filter, and the image of the selected content may be displayed on the display **108**.

2-4. Second Display Process at the Time of Selecting Genre Filter: Preference Program Guide

(158) Next, the second display process at the time of selecting a genre filter according to the second embodiment will be described with reference to FIG. **9**. FIG. **9** is a flowchart illustrating a flow of the second display process at the time of selecting a usual filter according to the first embodiment.

(159) In step S21 illustrated in FIG. **9**, the control unit **160** of the information processing device **100** determines whether the TV menu G2 at the time of selecting a genre filter is displayed. Since the processing similar to that in FIG. **9** is performed except for the process in step S21, the description thereof will be omitted. However, the “usual program guide” in steps S25, S28, and S30 illustrated in FIG. **9** is changed to the “preference program guide”.

(160) According to the second display process, it is possible to present the user with the selection candidate content that is highly likely to be desired by the user with the preference program guide G6, and thus, the user can avoid a complicated operation of selecting a channel and easily select and view the desired content. In addition, since it is possible to smoothly switch between the preference program guide G6 and the TV menu G2 at the time of selecting a usual filter, it is possible to improve user convenience.

2-5. Recommendation Display of Preference Program Guide

(161) Next, recommendation display of a preference program guide according to the second embodiment will be described with reference to FIG. 17. FIG. 17 is a second diagram for describing a second display screen (preference program guide) at the time of selecting a genre filter according to the second embodiment.

(162) For example, the search control unit **162** searches for a content (program) that the user is likely to be interested in by using big data (huge data group) or the like obtained from the outside via a network based on the preference information about the user. The display control unit **166** adds and recommends the contents obtained by the search by the search control unit **162** to the preference program guide **G6**.

(163) For example, in a case where an original author name as a favorite original author is included in the preference information, another author (for example, an author whose user does not know) related to the original author is recommended from the big data. Specifically, as illustrated in FIG. 17, in the preference program guide **G6**, “XXXX” that is a favorite original author is displayed, and contents related to “YYYY” that is a writer (for example, an author whose works created are close in content) close to the original author “XXXX” is displayed. As a result, it is possible to recommend an author that the user is likely to be interested in (likely to like) and to provide the content of another author (original author) to a user who views only the content of a specific original author. At this time, instead of providing a content that the user is not interested in, the content that the user is likely to be interested in is presented, and thus, it is possible to widen a selection range for selecting a content desired by the user and improve convenience of the user. Note that, even in a case where an entertainer name as a favorite entertainer is included in the preference information, or the like, similar processing can be performed.

(164) Furthermore, for example, in a case where an original author name as a favorite original author is included in the preference information, a keyword (for example, reversal, difficult situation, and the like) that is frequently included in the description of the drama of the original author may be obtained from the big data, and the content corresponding to another drama described by the obtained keyword may be recommended. In a case where a movie name as a favorite movie is included in the preference information, a keyword (for example, space, physics, and the like) largely included in the description of the movie may be obtained, and the content corresponding to another movie described by the obtained keyword may be recommended. Furthermore, in a case where an entertainer name as a favorite entertainer is included in the preference information, a face image of the stored entertainer may be analyzed and compared with a face image of another entertainer, an entertainer resembling the stored entertainer may be identified, and the content in which the identified entertainer appears may be recommended. In addition, for example, related information including an illustration, a photograph, or an explanatory sentence, such as an image of a recommended entertainer or a biographic introduction of a writer, may be displayed in the preference program guide **G6** in association with a plurality of contents. Furthermore, a GUI for presenting a screen displaying information about a recommended writer or entertainer may be displayed in the preference program guide **G6** in association with a plurality of contents. Displaying the GUI in association with the plurality of contents may include displaying the GUI in association with a group of contents (for example, a group of contents sorted by the vertical bar illustrated in FIG. 17) displayed in association with a writer or an entertainer on the preference program guide **G6**.

(165) Note that the content obtained by the search is added to the preference program guide **G6** and recommended as described above, but the present invention is not limited to this, and for example, the content may be added to the usual program guide **G3** and recommended. In addition, the TV menu **G2** may include recommended content corresponding to the preference program guide **G6** or the usual program guide **G3**.

2-6. Summary of Second Embodiment

(166) As described above, according to the second embodiment, the control unit **160** performs the

process of outputting the TV menu G2, which is an example of the second display image corresponding to the content selection candidate information (for example, the selection candidate content) according to the category information, based on the user selection corresponding to the category information (for example, various filters). As a result, since it is possible to present the user with the selection candidate content according to the category information that is highly likely to be desired by the user, the user can easily select and view the desired content.

(167) In addition, in the TV menu G2 at the time of selecting a genre filter, content selection candidates (selection candidate contents) are disposed in an order based on the content selection information. As a result, the content selection candidates are disposed in an order suitable for the user and provided to the user, so that the user can easily select the desired content. As a result, the user can easily view the desired content.

(168) In addition, the control unit 160 performs a process of switching from an output of the display content and the TV menu G2 at the time of selecting a usual filter to an output for displaying the preference program guide G6, which is an example of the third display image corresponding to the content information interface, on the entire screen based on the user operation for the TV menu G2 at the time of selecting the usual filter. As a result, it is possible to provide a content selection candidate suitable for the user with the preference program guide G6, so that the user can easily select the desired content. As a result, the user can easily view the desired content.

(169) In addition, in the preference program guide G6, content selection candidates (for example, selection candidate contents) are disposed in an order based on the content selection information. As a result, the content selection candidates are disposed in an order suitable for the user and provided to the user, so that the user can easily select the desired content. As a result, the user can easily view the desired content.

(170) In addition, in the preference program guide G6, selection candidates are added and recommended based on the content selection information. As a result, since it is possible to present the content that the user is likely to be interested in, it is possible to increase a selection range for selecting the content desired by the user and to improve convenience of the user.

3. Other Embodiments

3-1. Various Settings by Control Panel

(171) Next, various settings by a control panel according to other embodiments will be described.

(172) The control panel may have a user interface image for standard definition (SD) program non-display setting. By default, the setting is OFF. When the SD program non-display setting is set to ON, the SD program is not displayed in the TV menu G2. As a result, it is possible to suppress simultaneous display of programs having different resolutions of high definition (HD) or SD but the same program content. That is, one of the programs having different resolutions but the same content may be deleted from the TV menu.

(173) Note that not only one of the programs having different resolutions but the same content but also a program (content) not desired by many people may be deleted from the TV menu. In addition, an unsubscribed program, a data broadcast program, a program with a long program promotion, a program that cannot be viewed (program with no signal), a program in languages that the user cannot enjoy or understand (particularly, multi-language countries), and a program that cannot be viewed in a predetermined time zone (unique to the United States) may be deleted. Alternatively, a plurality of elements may be selectively displayed, such as which of subtitle and voice dubbing is displayed in the TV menu.

(174) As described above, since the user can perform various settings, the TV menu G2 at the time of selecting a usual filter and the TV menu G2 at the time of selecting a genre filter can be customized according to the user's preference. As a result, it is easier for the user to select the content desired by the user, so that the user can view the content desired by the user more reliably, and, as a result, the convenience of the user can be improved.

3-2. Selectable Display of Usual Filter

(175) Next, the selectable display of the usual filter will be described with reference to FIG. 18. FIG. 18 is a diagram for describing the selectable display of a usual filter according to another embodiment.

(176) In a case where the information amount of the content selection information (for example, the operation log) is equal to or less than the predetermined amount, as illustrated in the screen A1 of FIG. 18, the display control unit 166 performs a process of not including the “usual program”, which is the filter name of the usual filter, in the initial setting TV menu (the TV menu other than a TV menu at the time of selecting a usual filter) G2. On the other hand, when the information amount of the content selection information exceeds the predetermined amount, as illustrated in the screen A2 in FIG. 18, the display control unit 166 performs a process of automatically including the “usual program” in the initial setting TV menu G2.

(177) Furthermore, in a case where the “usual program” is selected for the first time on the screen A2 in FIG. 18, as illustrated in a screen A3 in FIG. 18, an image that introduces the function of the usual filter may be displayed, or the function of the usual filter may be introduced by voice. Note that, in a case where the information amount of the content selection information is equal to or less than the predetermined amount, in order to store the content selection information, an image prompting to store the content selection information may be displayed, or a voice prompting to store the content selection information may be output.

(178) As described above, in the initial setting the TV menu G2, when the information amount of the content selection information is equal to or less than the predetermined amount, the selectable category information (for example, the “usual program” of the usual filter) for selecting the usual filter is not disposed, and when the information amount of the content selection information exceeds the predetermined amount, the selectable category information for selecting the usual filter is disposed. As a result, when the information amount of the content selection information exceeds the predetermined amount, the user can select the usual filter, so that the processing based on the content selection information having a sufficient information amount is executed. That is, it is possible to determine whether to display an image of selectable category information corresponding to a usual filter based on the usage condition of the device. Therefore, since the content desired by the user can be more reliably provided, the user can easily view the desired content.

(179) Note that, as described above, the display/non-display of the selectable category information for selecting a usual filter (outputting the TV menu G2 at the time of selecting a usual filter) is controlled, but the present invention is not limited to this, and for example, the display/non-display of the selectable category information for selecting a genre filter (outputting the TV menu G2 at the time of selecting a genre filter) may be similarly controlled.

3-3. Customization of Usual Program Guide

(180) Next, customization of a usual program guide according to another embodiment will be described with reference to FIGS. 19 to 21. FIG. 19 is a first diagram for describing customization of a usual program guide according to another embodiment. FIG. 20 is a second diagram for describing customization of a usual program guide according to another embodiment. FIG. 21 is a third diagram for describing customization of a usual program guide according to another embodiment. Note that FIGS. 19 to 21 illustrate part of a usual program guide.

(181) Sometimes the user is not satisfied with the usual program guide G3 (for example, a list of frequently viewed channels and programs). In this case, it may be possible to customize the usual program guide G3 according to the input operation by the user. For example, the display control unit 166 performs a process of customizing the usual program guide G3 according to the input operation. Note that the display control unit 166 may perform a process of customizing the preference program guide G6 other than the usual program guide G3. In addition, a result of customization with respect to the usual program guide G3 and the preference program guide G6 may be reflected in presentation of the selection candidate content in the TV menu G2.

(182) As a case of customization, for example, there is a case where it is necessary to delete the content. The case where the deletion is necessary is a case where a channel or a program that is not desired to be displayed is displayed. Examples of the channel or the program that is not desired to be displayed include a channel or a program that does not want the family to watch, a channel or a program that is not interesting after being viewed for a trial, and a channel or a program whose subscription has been cancelled. In addition, there is a case where rearrangement of the content is necessary. As a case where rearrangement is necessary, there is a case where there is a favorite channel or a favorite program although it is not often watched. For example, since a channel or a program that the user does not really watch but the user likes is present at the bottom of a usual program guide, and difficult to find, the user may want to move the content to the top. In addition, there is a case where it is necessary to add the content. As a case where it is necessary to add the content, for example, there is a case where a channel or program that has not been watched for a while (has not been watched recently) disappears from a usual program guide.

(183) As illustrated in FIG. 19, in a case where the user displays a customized screen of a program guide in the usual program guide G3, a pin is displayed on each channel, and when the pin (pin image) of a dynamic channel is pressed, the dynamic channel at which the pin is pressed is a newly fixed channel. When the usual program guide G3 is opened next time, the priority of the newly fixed channel is higher than that of the dynamic channel, and it moves above the dynamic channel, for example, to the top of the usual program guide G3.

(184) In the usual program guide G3, there are a fixed channel (fixed CH) and a dynamic channel (dynamic CH), and the fixed channel and the dynamic channel can be switched. The fixed channel is a channel (content) added or set by the user. The dynamic channel is a channel (content) that dynamically changes in the logic of the "usual program". In the usual program guide G3, for example, a UI image (for example, a UI image having a different color) that enables determination of whether a channel is the fixed channel or the dynamic channel is displayed.

(185) As illustrated in FIG. 20, when a pin of a fixed channel is pressed in the usual program guide G3, the fixed channel at which the pin is pressed is a newly dynamic channel. When the usual program guide G3 is opened next time, the new dynamic channel is lower than the fixed channel, and moves under the fixed channel, for example, under the fixed channel and to the top of the dynamic channel group.

(186) In addition, as illustrated in FIG. 21 (upper stage), in the usual program guide G3, for example, when a deletion operation is performed using a remote controller in a state where a dynamic channel is selected, the selected dynamic channel is deleted from the usual program guide G3. Note that the fixed channel may be deleted by a similar operation.

(187) As illustrated in FIG. 21 (lower part), in a filter other than the usual filter, for example, when an addition operation is performed using a remote controller in a state where a channel (content) is selected, the selected channel is added to the usual program guide G3. At this time, the selected channel may also be added to favorites.

(188) As described above, in the usual program guide G3 (or preference program guide G6), the order of the selection candidate contents, which is an example of the selection candidates, is changed by the user operation. In addition, the selection candidate content is added or deleted by a user operation. That is, since the user can customize the usual program guide G3 (or the preference program guide G6) by sorting, deleting, or adding, for example, the selection candidate contents in the usual program guide G3 (or the preference program guide G6), the user can create the usual program guide G3 (or the preference program guide G6) based on the user's preference. As a result, it is easier for the user to select the content desired by the user, so that the user can view the content desired by the user more reliably, and, as a result, the convenience of the user can be improved.

3-4. Highlight Display of TV Menu

(189) Next, the highlight display of a TV menu according to another embodiment will be described with reference to FIGS. 22 and 23. FIG. 22 is a first diagram for describing a highlight display of a

TV menu according to another embodiment. FIG. 23 is a second diagram for describing a highlight display of a TV menu according to another embodiment.

(190) In a case where a plurality of broadcast standards is juxtaposed, there is a case where the same content (program) is broadcast in a new high-quality service with higher utilization efficiency of the radio band and an old service with reduced allocation of the radio band. In addition, according to an operation situation, there is a program transmitted only by a new broadcast standard while there is a program transmitted only by an old broadcast standard. Since it is possible to present the user with the content by distinguishing them by the highlight display, it is easier for the user to select the content desired by the user, and it is possible to view the content desired by the user more reliably. As a result, user convenience can be improved.

(191) For example, as illustrated in FIG. 22, an icon (for example, “standard 1”) G7 indicating the broadcast standard is displayed in an item image corresponding to the content that is required to clearly indicate the broadcast standard among the item images of the TV menu G2. The content that is required to clearly indicate the broadcast standard is, for example, a content whose content is the same but that is broadcast in a different broadcast standard. Furthermore, other than the icon, for example, as illustrated in FIG. 23, a color of the edge of the item image corresponding to the content which is required to clearly indicate the broadcast standard among the respective item images of the TV menu G2 may be changed to a conspicuous color (for example, red) and highlighted.

(192) As described above, in the TV menu G2, the item image corresponding to the specific content is highlighted. For example, the item image corresponding to the content broadcasted in a specific broadcast standard according to the difference in the broadcast standards is highlighted. In this manner, it is possible to present the user with the content by differentiate the broadcast standard of the content. As a result, it is easier for the user to select the content desired by the user, so that the user can view the content desired by the user more reliably, and, as a result, the convenience of the user can be improved.

(193) Note that, in a case where the same content can be acquired in a plurality of broadcast standards at the same time, the content based on a newer broadcast standard (which transmits high-quality content) may be preferentially selected. This may be performed in both cases of selecting a channel with a usual filter and selecting a channel normally. For example, the high-quality content is a content corresponding to a high resolution (4K/8K or the like), a high bit depth (high bit rate), a high luminance (HDR or the like), a high frame rate (HFR or the like), and a high color gamut. In addition, a filter name capable of selecting a first broadcast standard of a second broadcast standard older than the first broadcast standard may be added as a filter name displayed in the TV menu G2.

(194) Furthermore, in a case where the same content can be acquired in a plurality of broadcast standards at the same time, a description or an icon indicating that the program is transmitted in a plurality of standards may be displayed on the TV menu G2 (for example, a mini-program guide) or the EPG, or a method of displaying the TV menu G2 or the EPG may be changed according to the standard. The display of the description and the icon, and a change in display method may be indicated only for the content with a standard different from the currently selected broadcast standard, or may indicated only for the content with a standard corresponding to a newer standard (transmitting high-quality content).

(195) Furthermore, whether it is possible to acquire the same content in a plurality of broadcast standards at the same time may be determined based on broadcast metadata transmitted in one or both of a broadcast wave (broadcast stream) of a first (new or old) broadcast standard and a broadcast stream of a second (old or new) broadcast standard. The broadcast metadata included in the broadcast stream of the first broadcast standard may include information (for example, an identifier or a flag) indicating whether the same content is transmitted in the broadcast stream of the second broadcast standard (the first and the second may be reversed). The broadcast metadata is information used for broadcast scanning, and the receiver may create a list of receivable channel

information according to the broadcast metadata. Furthermore, the list may be displayed on the screen as, for example, an EPG or a TV menu.

(196) Here, the broadcast metadata may be distributed by communication. That is, the communication metadata is completely or almost the same as the broadcast metadata. As the communication metadata, data obtained by adding information about the broadcast content to a description related to the Internet content (communication content) may be distributed by communication. This may be completely different from the broadcast metadata. The broadcast content(s) and the communication content(s) may be compared, and the highest quality content may be preferentially selected. This may be performed in either case of tuning to a usual channel or normal tuning. In addition, the broadcast content may be prioritized in consideration of a delay problem or the like, or a frequently used source may be preferentially selected. The high-quality content may be determined based on only one of a resolution, a bit depth (bit rate), a luminance, a frame rate, and a color gamut, may be determined based on a plurality of them, or may be determined based on a delay, broadcast reception sensitivity, a communication line speed, or the like.

(197) Further, each communication content may be designated by an Internet content identifier. In addition, each communication content may include a broadcast recording content and a communication-downloaded content. After the same contents are collectively displayed, an icon indicating a distribution source, an icon indicating contents in broadcast, and an icon indicating recorded contents may be displayed to indicate from which source the contents can be acquired.

(198) Note that, in each of the above embodiments, the description has been mainly made based on the moving image content of the broadcast program or the like as a content to be searched for and executed, but the content type is not restricted as long as the content can be executed by a device or equipment realized by the present disclosure. For example, the content to be played back may be a voice content (for example, music), and the music is played back by an application executable by the information processing device **100**. Furthermore, the target to be searched for based on the priority may be a voice content or an application by which the voice content is played back. In this case, one piece of music or an application is selected based on a procedure similar to any of the above-described embodiments or a combination thereof, music or applications that has not been selected is set as selection candidates, and a TV menu corresponding to the selection candidates is displayed on the display **108**. For example, an album art image (jacket image) of music as the selected content, a user interface image of an application, and the like are displayed on the display **108**. The channel described in each of the above embodiments may be a channel such as a radio or a music distribution station that broadcasts a voice content such as music. The same procedure is applied to other types of contents such as a still image content and web page.

(199) Note that, for the usual program guide **G3** and the preference program guide **G6** described in each of the above embodiments, the full-screen program guide may be displayed in a state in which the usual filter or each genre filter is selected even when the program guide display is requested based on the pressing operation of the program guide button of the remote controller or the voice input. In addition, the program guide may be displayed in a state in which the usual filter or each genre filter is selected only at the time of transition from the TV menu. In a case where the full-screen program guide is displayed without going through the TV menu **G2**, the full-screen program guide may be displayed in a state in which the filter of the content source (for example, terrestrial digital broadcasting) being viewed is selected. Whether to preferentially display the usual program guide **G3** or preference program guide **G6** when not going through the TV menu **G2** may be selected by the user on a setting screen or the like illustrated in FIG. **11**. In addition, in order to preferentially display the usual program guide **G3** and the preference program guide **G6**, a dedicated button may be provided on the remote controller, or a predetermined request operation (for example, continuous pressing of a program guide button or utterance such as “open the usual program guide”) may be received using a remote controller input, a voice input, a gesture input, or

the like.

(200) Although the embodiments have been described in detail with reference to the accompanying drawings, the present technology is not limited to such examples. Respective embodiments may be implemented in combination. It is obvious that those skilled in the art in the technical field of the present disclosure can find various revisions and modifications within the scope of a technical concept described in claims, and it should be understood that these revisions and modifications will also be naturally come under the technical scope of the present disclosure.

(201) For example, it is also possible to create a computer program for causing hardware such as a CPU, a ROM, and a RAM built in the information processing device **100** to exhibit the functions of the information processing device **100**. Furthermore, a computer-readable storage medium storing the computer program is also provided.

(202) Note that the effects described in the present specification are merely illustrative or exemplified effects, and are not limitative. That is, the technology according to the present disclosure can accomplish other effects apparent to those skilled in the art from the description of the present specification, in addition to or instead of the effects described above.

4. Supplementary Note

(203) The present technology may also be configured as below. (1) An information processing device comprising: a control unit configured to perform a process of acquiring content selection information about a user, a process of determining a display content and content selection candidate information based on the content selection information, and a process of outputting the display content and a first display image corresponding to the selection candidate information. (2) The information processing device according to the (1), wherein the first display image is created based on one or both of a service and a content used by a user within a predetermined period. (3) The information processing device according to the (1) or (2), wherein the first display image changes according to one or both of a time zone and a day of the week. (4) The information processing device according to any one of the (1) to (3), wherein the first display image includes a plurality of pieces of selectable category information corresponding to a content acquisition means or a content type. (5) The information processing device according to the (4), wherein the control unit performs a process of outputting a second display image corresponding to selection candidate information about the content according to the category information based on user selection corresponding to the category information. (6) The information processing device according to the (5), wherein content selection candidates are disposed in an order based on the content selection information in the second display image. (7) The information processing device according to any one of the (1) to (6), wherein content selection candidates are disposed in an order based on the content selection information in the first display image. (8) The information processing device according to any one of the (1) to (7), wherein the control unit performs a process of switching from an output of the display content and the first display image to an output for displaying a third display image corresponding to a content information interface on an entire screen based on a user operation on the first display image. (9) The information processing device according to the (8), wherein the first display image and the third display image include selectable category information at corresponding positions. (10) The information processing device according to the (8) or (9), wherein content selection candidates are disposed in an order based on the content selection information in the third display image. (11) The information processing device according to any one of the (8) to (10), wherein the first display image or the third display image includes lower category information related to an element included in a content. (12) The information processing device according to the (11), wherein the first display image or the third display image includes at least one of an illustration, a photograph, or an explanatory sentence related to an element included in a content. (13) The information processing device according to any one of the (10) to (12), wherein the selection candidate is added and recommended based on the content selection information in the first display image or the third display image. (14) The information processing device according to

any one of the (10) to (13), wherein the selection candidate is changed in an order, added, or deleted by a user operation in the third display image. (15) The information processing device according to any one of the (1) to (14), wherein the control unit determines whether to display the display content or the first display image based on a usage condition of the information processing device. (16) The information processing device according to the (15), wherein the control unit determines whether to display the display content or the first display image based on an elapsed time from a previous activation of the information processing device. (17) The information processing device according to any one of the (1) to (16), wherein the control unit performs a process of outputting setting information for selecting to display a display content selected based on the content selection information at a time of activation, and a process of displaying a display content selected based on the content selection information at a time of activation based on user selection for the setting information. (18) The information processing device according to any one of the (1) to (17), wherein the control unit performs a process of acquiring a content output instruction related to the content selection information, and a process of determining the display content and the selection candidate information when the content output instruction is acquired. (19) The information processing device according to the (18), wherein the content output instruction is an utterance of a user, and the control unit performs a process of acquiring, based on an utterance of the user, content selection information corresponding to the user who has made the utterance, and a process of determining the display content and the selection candidate information based on the acquired content selection information. (20) An information processing method executed by a computer, the method comprising: a process of acquiring content selection information about a user; a process of determining a display content and content selection candidate information based on the content selection information; and a process of outputting the display content and a first display image corresponding to the selection candidate information.

REFERENCE SIGNS LIST

(204) **100** INFORMATION PROCESSING DEVICE **102** MICROPHONE **103** INPUT DEVICE **104** COMMUNICATION I/F (INTERFACE) **105** MEMORY **106** TUNER **107** DECODER **108** DISPLAY **112** PROCESSOR **120** VOICE ACQUISITION UNIT **130** VOICE RECOGNITION UNIT **140** FEATURE DATABASE (DB) **150** APPLICATION UNIT **160** CONTROL UNIT **162** SEARCH CONTROL UNIT **164** SELECTION CONTROL UNIT **166** DISPLAY CONTROL UNIT **172** HISTORY DB (DATABASE) **174** PREFERENCE DB (DATABASE) **G1** IMAGE OF DISPLAY CONTENT **G2** TV MENU **G3** USUAL PROGRAM GUIDE **G4** SETTING IMAGE **G5** SETTING IMAGE **G6** PREFERENCE PROGRAM GUIDE

Claims

1. An information processing device comprising: circuitry configured to; acquire content selection information about a user, determine a display content and content selection candidate information based on the content selection information, and output the display content and a first display image corresponding to the content selection candidate information, wherein the circuitry is further configured to determine whether to display the display content or the first display image based on an elapsed time from a previous activation of the information processing device.
2. The information processing device according to claim 1, wherein the first display image is created based on one or both of a service and a content used by the user within a predetermined period.
3. The information processing device according to claim 1, wherein the first display image changes according to one or both of a time zone and a day of the week.
4. The information processing device according to claim 1, wherein the first display image includes a plurality of pieces of selectable category information corresponding to a content acquisition means or a content type.

5. The information processing device according to claim 4, wherein the circuitry further configured to output a second display image corresponding to the content selection candidate information according to the selectable category information based on user selection corresponding to the selectable category information.
6. The information processing device according to claim 5, wherein content selection candidates are disposed in an order based on the content selection information in the second display image.
7. The information processing device according to claim 1, wherein content selection candidates are disposed in an order based on the content selection information in the first display image.
8. The information processing device according to claim 1, wherein the circuitry further configured to switch from an output of the display content and the first display image to an output for displaying a third display image corresponding to a content information interface on an entire screen based on a user operation on the first display image.
9. The information processing device according to claim 8, wherein the first display image and the third display image include selectable category information at corresponding positions.
10. The information processing device according to claim 8, wherein content selection candidates are disposed in an order based on the content selection information in the third display image.
11. The information processing device according to claim 8, wherein the first display image or the third display image includes lower category information related to a first element included in a first content.
12. The information processing device according to claim 11, wherein the first display image or the third display image includes at least one of an illustration, a photograph, or an explanatory sentence related to a second element included in a second content.
13. The information processing device according to claim 10, wherein a selection candidate is added and recommended based on the content selection information in the first display image or the third display image.
14. The information processing device according to claim 10, wherein a selection candidate is changed in an order, added, or deleted by some user operation in the third display image.
15. The information processing device according to claim 1, wherein the circuitry further configured to determine whether to display the display content or the first display image based on a usage condition of the information processing device.
16. The information processing device according to claim 1, wherein the first display image is created based on a channel used by the user within a predetermined period.
17. The information processing device according to claim 1, wherein the circuitry further configured to: output setting information for selecting to display some display content selected based on the content selection information at a time of activation, and display other display content selected based on the content selection information at some time of activation based on user selection for the setting information.
18. The information processing device according to claim 1, wherein the circuitry further configured to: acquire a content output instruction related to the content selection information, and determine the display content and the content selection candidate information when the content output instruction is acquired.
19. The information processing device according to claim 18, wherein the content output instruction is an utterance of the user, and the circuitry further configured to: acquire, based on an utterance of the user, content selection information corresponding to the user who has made the utterance, and determine the display content and the content selection candidate information based on the acquired content selection information.
20. An information processing method comprising: acquiring content selection information about a user; determining a display content and content selection candidate information based on the content selection information; outputting the display content and a first display image corresponding to the content selection candidate information, determining whether to display the

display content or the first display image based on an elapsed time from a previous activation event.
